

MID

Course: Data Structures (CS101)

Set: B | Time: 2 Hours | Max Marks: 60 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
attend all the questions

SECTION A: MCQs

1. Which of the following is NOT listed as a desired property that having an acceptable system involves for stakeholders in Section 1.1? **[5 Marks]**

- a) Performance
- b) Marketability
- c) Security
- d) Modifiability

2. According to the definition provided in Section 1.2, an architecture is primarily an abstraction of a system that suppresses which of the following? **[5 Marks]**

- a) Details of all software elements within the system.
- b) Details of elements that do not affect how they use, are used by, relate to, or interact with other elements.
- c) Details of externally visible properties of elements.
- d) Details of all relationships between elements.

3. Which statement accurately describes the relationship between "Architectural Pattern" and "Architectural Style" as presented in the text? **[5 Marks]**

- a) An architectural pattern is a foundational building block for an architectural style.
- b) An architectural pattern in software is also known as an architectural style.
- c) Architectural styles are a more general concept, while patterns are specific instances.
- d) Architectural patterns and styles are entirely distinct concepts with no overlap.

4. In the context of "Module Views," what specific property is primarily used to identify a module's role in the overall system with more certainty than its name? **[5 Marks]**

- a) Name
- b) Implementation information

- c) Responsibilities
- d) Revision history

5. The Attribute-Driven Design (ADD) approach to defining a software architecture bases its decomposition process primarily on what? **[5 Marks]**

- a) The set of business and technical decisions made early in the life cycle.
- b) The experience of the architects and the developing organization.
- c) The quality attributes the software has to fulfill.
- d) The specific functional requirements expressed as use cases.

6. Which of the following is NOT categorized as a "System Quality" attribute in the provided text's "Quality Attributes" section (1.5), but rather belongs to a different category? **[5 Marks]**

- a) Modifiability
- b) Usability
- c) Time to Market
- d) Security

SECTION B: SHORT ANSWERS

7. According to the text, how do "System stakeholders" influence software architectures, and what three crucial responsibilities must architects fulfill when engaging these stakeholders early in the life cycle? **[6 Marks]**

8. Based on the definition of software architecture provided, explain two key implications concerning the nature of software elements and the abstraction of a system. **[6 Marks]**

9. Differentiate between an Architectural Pattern, a Reference Model, and a Reference Architecture. Provide a defining characteristic for each and state whether each is considered an "architecture" itself. **[6 Marks]**

10. Describe three typical properties associated with elements (components or connectors) in a Component-and-Connector (C&C) view, and for each, explain its purpose or how it supports analysis. **[6 Marks]**

11. Architecture documentation fundamentally serves three uses. Explain each of these uses in your own words, drawing directly from the text. **[6 Marks]**